# MICHAEL JEFFERSON

## FORWARD-DEPLOYED ENGINEER & FULL-STACK SOLUTIONS BUILDER

Vacaville, CA (US Remote & SF Bay Area Hybrid) | Mjefferson@gm.slc.edu | michael-jefferson-portfolio.vercel.app | github.com/jeffersonlegacy | axion-tax-beyond.vercel.app

Client-focused Full-Stack and Applied AI Engineer who works directly with business owners to scope operational bottlenecks, prototype custom **document intake, voice discovery, and real-time WebRTC applications**, and deploy working software in **TypeScript, React, Next.js, and Node.js**. Pairs software execution (2024–Present) with 7 years of quantitative instruction and clear stakeholder communication.

## TECHNICAL STACK

**Forward-Deployed & Solutions Engineering:** Stakeholder Discovery & Scoping, Rapid Prototype-to-Deployment Execution, Executive PDF Deliverable Automation, Bilingual (EN/ES) Client Workflows, A2P 10DLC Compliance, Technical Onboarding

**AI, Document Workflows & Agentic Tooling:** Gemini API (Pro / Flash / Vision), Vertex AI, OpenAI SDK & Codex, Document Triage & OCR Pipelines (Tesseract.js, ZXing, Vision LLMs), Ollama (Local LLM Inference), Multi-Agent Orchestration (OpenClaw), Structured JSON Schema Contracts, Context Routing (Agent Desk)

**Frontend & Interactive UX:** TypeScript, JavaScript (ES6+/ESM), Next.js (App Router, SSR/RSC), React 19, Vite, Tailwind CSS, Chart.js, Web Audio API, Remotion (Programmatic Video), Tldraw Canvas

**Backend, Real-Time & Cloud:** Node.js (ESM), Express 5, Firebase (Firestore, Realtime Database, Auth, Storage), LiveKit (WebRTC & RBAC JWTs), Twilio 10DLC SMS API & Webhooks, Resend, CloudConvert API, REST APIs

## SOFTWARE & APPLIED AI ENGINEERING EXPERIENCE

**Forward-Deployed Full-Stack & Solutions Engineer — Jefferson Legacy Systems (Independent Client & Product Engineering) |** Vacaville, CA | 2024 – Present

- **Axion Tax & Beyond (axion-tax — Owner-Reviewed Client Prototype):** Built and deployed a full-stack tax intake portal (**Vite + React 19 SPA on Vercel** and **Node.js ESM Express 5 API**) featuring categorized upload lanes, client-side **SHA-256 file fingerprinting and filename-based document triage**, Schedule C & Write-Off Discovery wizards, and bilingual (EN/ES) Twilio SMS + Resend Email alerts.
- **Automated Visual QA & PDF Deliverable Pipeline:** Engineered a headless **Playwright** viewport capture workflow (output/playwright/) paired with a custom **Python PDF synthesis script (build_axion_pitch_deck.py)** to programmatically compile live application states into an executive pitch deck.
- **Tasty Cuts & More (tasty-cuts-and-more — Local Delivery Commerce & Owner Discovery Prototype):** Built a household freezer-pack delivery storefront and **48-question Web Audio voice discovery intake tool** in **React 19, Vite 8, and Firebase Hosting** with local promo validation (TASTY10 / FILLTHEFREEZER) and structured JSON/text brief export.
- **Tower Hour (tower-hour — Live Social Commerce & Multimodal Scanner Platform):** Developed a live-event shopping and product scanner application in **Next.js 16 App Router, TypeScript, Tailwind CSS, and Firebase** (Firestore + RTDB) across a 373-SKU catalog.
- **WebRTC RBAC & Twilio 10DLC SMS State Machine:** Implemented cryptographically signed **LiveKit RBAC JWTs** isolating broadcaster publish permissions from viewer subscribe tokens, alongside a bidirectional **Twilio 10DLC webhook pipeline** and **Sentry** observability.
- **Multimodal Product Scanner & Local Remotion Worker:** Combined **Tesseract.js, ZXing, and Vertex AI / Gemini** (/api/scan) to map photographed household products to catalog SKU replacements, and built a local **Remotion** worker for zero-cloud-cost video rendering.
- **Jefferson Super Reader 3000 (Document Vision & RSVP Instrument):** Built an asynchronous document conversion and reading pipeline integrating **CloudConvert** and **Gemini Vision APIs** with multi-word chunking, progressive WPM ramp, and a client-side **Optimal Recognition Point (ORP)** RSVP engine.
- **Mike Mac Mesh & Coach Mike AI:** Provisioned a headless **Debian Linux** inference server (tuned **zram** swap + **Ollama**) bridged via **2.5GbE LAN / Tailscale** to an M2 Mac mini with file-first context routing (agentdesk), alongside an interactive **Chart.js** role-alignment coaching visualizer.

**Founder & STEM Systems Instructor — Tower Tutoring |** Vacaville, CA | 2023 – Present

- Built an interactive collaborative tutoring web platform (**React, Vite, LiveKit WebRTC, Tldraw canvas, OpenAI SDK, Firebase**) and delivered 1-on-1 quantitative coaching, applying cognitive load principles to human-in-the-loop educational tools.

## INSTRUCTIONAL & STAKEHOLDER COMMUNICATION EXPERIENCE

**Mathematics Instructor & Education Specialist — Public School Districts (EOLA / VUSD / TUSD) |** Solano & Alameda Counties, CA | 2018 – 2025

- Led 7 years of classroom Algebra and Geometry instruction and designed data-driven STEM curricula, translating complex quantitative frameworks into clear mental models for diverse students, parents, and administrative stakeholders.

## EDUCATION

**Sarah Lawrence College** — Bronxville, NY | **Bachelor of Arts** (Psychology & Neuroscience Concentration; Kinetic Sculpture & Physical Computing)

**LaGuardia Community College** — Long Island City, NY | **Associate of Arts**